

Radar HRRP statistical recognition based on hypersphere model

Lan Du, Hongwei Liu*, Zheng Bao

National Laboratory of Radar Signal Processing, Xidian University, 2 Taibai Road, Xi'an Shaanxi 710071, China

Received 7 March 2007; received in revised form 22 September 2007; accepted 13 November 2007

Available online 21 November 2007

Abstract

The theoretical analysis and experimental results in this paper show that the independence assumption regarding elements in a radar high-resolution range profile (HRRP) sample, under which some statistical recognition methods were proposed, is not true. In addition, geometrically speaking, target HRRP samples spread on a unit hypersphere, since L_2 -normalized HRRP samples are applied to radar automatic target recognition (RATR) to deal with the amplitude-scale sensitivity problem. Therefore, this paper considers the two issues and proposes a novel statistical recognition method based on hypersphere model for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis. Compared with the conventional principal components analysis (PCA)-based subspace statistical recognition method, the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace can be fully utilized in this method without increasing computation complexity. The experimental results based on measured data show that our proposed method can greatly improve the recognition performance.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Radar automatic target recognition (RATR); High-resolution range profile (HRRP); Statistical recognition; Power transformation; Box–Cox transformation; Principal components analysis (PCA)

1. Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1–21]. In recent Refs. [9,10] discussed the properties of target HRRP; [11,12,18] studied some time-shift invariant features of HRRP; [13] discussed the variation of recognition performance with signal bandwidth; [14] proposed a new feature selection method for HRRP; [15] applied hidden Markov model (HMM) to radar HRRP target recognition and [16,17] proposed some advanced classifiers for HRRP-based RATR. However, there has been few literatures working on statistical modeling for radar target HRRPs, which is fundamental work for statistical recognition. By statistical recognition it was meant that the class membership of a

*Corresponding author Tel.: +86 29 88201448; fax: +86 29 88202260.

E-mail address: hwliu@xidian.edu.cn (H. Liu).

measurement HRRP $\mathbf{x}$ is determined based on the posterior probabilities of all class memberships $P(k|\mathbf{x})$ ($k = 1, 2, \dots, K$ denotes class membership). Several Refs. [19–21] showed that statistical recognition has the advantages of giving a measure of confidence for the class predictions, and the opportunity to combine the probabilities with additional information, such as intelligence reports, or possibly other sensor data. Furthermore, since a user needs to decide upon a course of action after classifying a target, the probabilities can be incorporated into a multilevel model that reflects the whole decision making process. For instance, by considering the expected posterior loss of decisions, the different costs involved in making a classification can be taken into account.

How to accurately describe HRRP's statistical characteristics is a challenging task in radar HRRP-based statistical recognition. Some statistical models [6,19–21] were proposed for HRRP samples under the assumption that the range cell elements in an HRRP sample are statistically independent, namely, the echo from a range resolution cell is statistically independent with that from another range resolution cell. Theoretical analysis and our experimental results based on measured data show that the independence assumption is not true. A statically correlated model is expected to describe the HRRP's statistical characteristic more accurately. Some Refs. [6,7,24] showed that the power transformation or the Box–Cox transformation with proper parameters can correct for the departures from Gaussian distribution to some extent. Thus, jointly multivariate Gaussian model can be used for statistical recognition based on power transformed HRRP samples, with which the difficulty in statistical modeling for HRRP samples can be greatly reduced.

As we know, the principal components analysis (PCA)-based subspace statistical recognition method can be used to analysis jointly multivariate Gaussian-distributed data. However, one issue has to be considered when HRRP is applied to statistical recognition. The amplitude-scales of HRRP samples cannot be used in HRRP-based RATR. This is what we call amplitude-scale sensitivity problem of radar HRRP samples. This comes from the fact that the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on. Thus, HRRP samples measured by different radars or under different conditions will have different amplitude-scales, and the measurement conditions of training samples and test samples are difficult to be kept exactly same in applications. For example, for a two-coordinate searching radar, the antenna gain along perpendicular axis varies widely, and it is difficult to know the exact gain of a target to scale its HRRP sample's amplitude. Therefore, amplitude L_2 -normalized HRRP samples are generally applied to RATR. Thus, different from general Gaussian-distributed data, geometrically speaking, power transformed HRRP samples spread on a unit hypersphere. This paper proposes a novel statistical recognition method based on hypersphere model for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis. Compared with the conventional PCA-based subspace method, the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace can be fully utilized in this method without increasing computation complexity. The experimental results based on measured data show that our proposed method can greatly improve the recognition performance.

This paper is organized as follows. In Section 2, based on the correlation analysis on HRRP samples and discussion about the power transformation and Box–Cox transformation, the conventional PCA-based subspace statistical recognition method for HRRP-based RATR is introduced. In Section 3, a new method which can fully utilize the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace is proposed. In Section 4, the main procedure of the statistical recognition methods for HRRP-based RATR is described. In Section 5, recognition experiments based on measured data are performed to compare the recognition performances of some independent statistical models, the conventional PCA-based subspace method and our proposed methods. Finally, conclusions are made in Section 6.

2. Conventional PCA-based subspace statistical recognition method for HRRP-based RATR

As discussed in Refs. [11,21], it is a prerequisite for HRRP-based RATR to deal with the target-aspect, time-shift and amplitude-scale sensitivity problems. To deal with the target-aspect sensitivity, a subset of HRRP samples from a target-aspect sector, within which the scatterers' motion through range cells

(MTRC) can be ignored, is defined to be an HRRP frame, which can be described by a statistical model [21]. To deal with the time-shift sensitivity, time-shift compensation techniques in ISAR imaging, such as the slide correlation processing or its modifications [22,23], can be used in both of the training phase and test phase [21]. For the amplitude-scale sensitivity, as discussed in Section 1, the L_2 amplitude normalization can be used as a preprocessing procedure. Thus, we discuss that how to accurately describe a time-shift compensated and amplitude L_2 -normalized HRRP frame's statistical characteristics in the following sections.

2.1. Statistical correlation property of HRRP samples

Generally, since the targets and their components are much larger than the wavelength of microwave radar, the electromagnetism characteristics of the targets can be described by the scattering center model [22,23], which is a first order approximation model without considering the multiple reflections between scatterers and has been proved to be a simple and efficient target model in synthetic aperture radar (SAR) imaging and inverse synthetic aperture radar (ISAR) imaging. According to this model, for high-resolution radar system, an aircraft-like target does not appear as a “point target” any longer, but consists of many scatterers separated in several range cells along the radar LOS. The scattering center model of a target will vary with the target-aspect, because the variation of target-aspect may cause some scatterers to remove from one range cell to another, which is referred to as scatterers' MTRC. The limitation of target-aspect change for avoiding scatterers' MTRC is given by [22,23]

$$\delta\varphi \leq (\delta\varphi)_{\text{MTRC}} \triangleq \frac{C}{2BL_x}, \quad (1)$$

where B , L_x and C are the bandwidth of the radar signal after range compression, the maximum target dimension in cross range and the speed of light, respectively. As the aforementioned discussion, the HRRP frames are divided just according to Eq. (1). Because the scatterers are localized immovably in each range resolution cell within an HRRP frame, of which the distribution is irrelevant to that in another cell, several Refs. [6,19–21] assumed that the echoes from different range resolution cells within an HRRP frame are independent.

Actually, this independence assumption is not true. There are several reasons:

- (a) According to the scattering center model, the echo from a range cell varies with the distance differences between the scatterers in this cell. When a target rotates, the scatterers from different range cells will rotate in the same way, which causes that the scatterers' distance differences in different cells dependently change; therefore, the statistical properties of the echoes from different cells are also dependent to some extent.
- (b) The returned echoes are multiplied by window functions before getting HRRP samples (e.g. the Hamming window was applied for the measured data in this paper) in order to compress the range sidelobe; therefore, the range resolution decreases and the echoes from two neighboring range cells (not those from the real range cells defined in Section 1) in an HRRP sample are not completely independent.
- (c) Different from what assumed in the scattering center model, some ISAR images using measured data show that the multiple reflections phenomena exist. When the multiple reflections phenomenon occurs between the scatterers in two range cells, the returned echoes from the two cells cannot be independent.
- (d) The similarity in geometrical structure and scattering property between different parts of a target can also cause the echoes from different range cells to be statistically correlated.

The correlation phenomena between the echoes from different range cells can be found in Fig. 1(a) and (b), which show the normalized covariance matrices of two HRRP frames from measured data (detailed introduction shown in Section 5.1). Therefore, this paper will make a detailed study on radar HRRP statistical recognition under the prerequisite that the elements in an HRRP sample are statistically correlated.

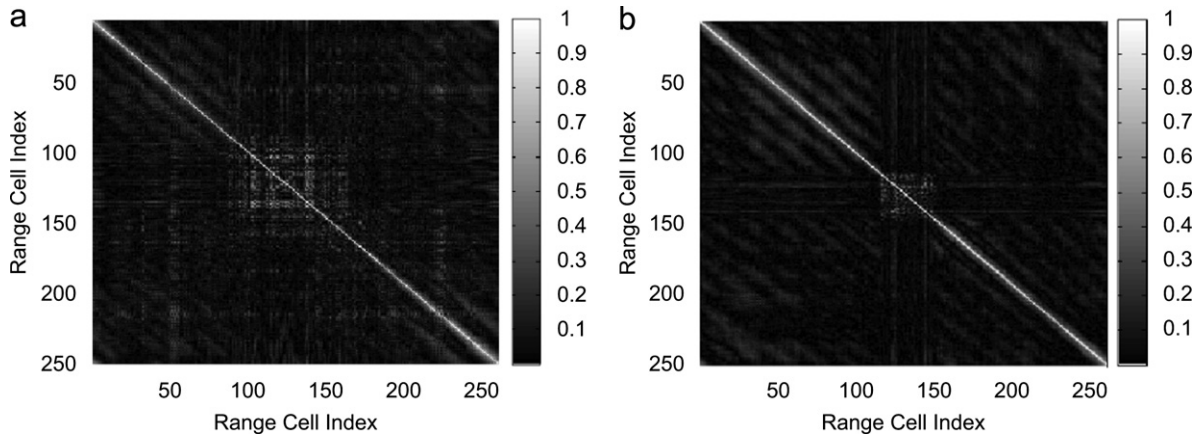


Fig. 1. Correlation analysis based on the normalized covariance matrices of two HRRP frames from measured data: (a) an HRRP frame of “Yak-42” and (b) an HRRP frame of “Cessna Citation S/II”.

2.2. Power transformation and Box–Cox transformation

The power transformation of x with parameter α is defined as

$$y = x^\alpha, \quad \alpha > 0 \quad (2)$$

and the Box–Cox transformation is defined as

$$y = \begin{cases} \frac{x^\alpha - 1}{\alpha} & 0 < \alpha \leq 1, \\ \log x & \alpha = 0. \end{cases} \quad (3)$$

For vectors $\mathbf{x}$, this transformation is applied to each element separately with the same α . Here α is a tuning parameter. When $\alpha = 1$, the power transformation has no influence on the original data, and the Box–Cox transformation is a linear transformation. Eqs. (2) and (3) require that all elements of vector $\mathbf{x}$ are positive. In RATR based on HRRP this will always be the case as the elements of $\mathbf{x}$ will have the dimension of amplitude. For generalizations of the Box–Cox transform for negative values we refer to Ref. [25].

It is well known that the performances of many techniques in pattern recognition (like a linear discriminant function and a nearest neighbor method) are optimal if Gaussian distribution is achieved. Moreover, in case of Gaussian distribution some techniques are more efficient and have lower error rates, for example in statistical recognition the parameter estimation for Gaussian model is much simpler than those for other models. Nevertheless, non-Gaussian distribution cases are encountered in many real world discrimination problems, e.g. the HRRP-based RATR. Refs. [6,7,24] showed that the power transformation and the Box–Cox transformation with proper parameters can correct for the departures from Gaussian distribution to some extent such that the techniques that assume or profit from Gaussian distribution can still be successful. Therefore, the power transformation or the Box–Cox transformation can be used as the preprocessing method for discrimination problems departing from Gaussian distribution. Nevertheless, these references only experimentally showed that the power transformation or the Box–Cox transformation can improve the recognition performance, but did not mathematically prove that the transformed data can strictly follow Gaussian distribution. Considering the correlation between elements in an HRRP sample, in this paper, we assume that the power transformed HRRP samples with proper power parameter approximate to follow jointly multivariate Gaussian distribution in HRRP-based RATR.

2.3. Main idea of conventional PCA-based subspace statistical recognition method for HRRP-based RATR

The PCA-based subspace statistical recognition method is a well-known statistical recognition method for jointly multivariate Gaussian-distributed data in pattern recognition area. In the following, we will introduce how to apply this method to HRRP-based RATR.

Let $\mathbf{X}_{kl} = \{\mathbf{x}_{kli} \in \mathbb{R}^d | i = 1, 2, \dots, N\}$ denote the l th ($l = 1, 2, \dots, L_k$) HRRP frame of target T_k ($k = 1, 2, \dots, K$), of which d denotes the sample vector dimensionality (usually d is large, e.g. $d = 256$ for our measured data.), N denotes the number of samples in an HRRP frame, and each d -dimensional sample $\mathbf{x}_{kli}$ are power transformed, range aligned and amplitude normalized (the detailed explanation about an HRRP frame, range alignment and amplitude normalization is shown in the first paragraph in Section 2 and in some Refs. [11,21]). Assume that the HRRP frame follows jointly multivariate Gaussian distribution, namely a sample from this frame $\mathbf{x} \sim \mathcal{N}(\mathbf{m}_{\mathbf{x}|kl}, \mathbf{C}_{\mathbf{x}|kl})$,

$$\mathbf{m}_{\mathbf{x}|kl} = E[\mathbf{x}] \approx \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli}, \quad (4)$$

$$\mathbf{C}_{\mathbf{x}|kl} = E[(\mathbf{x} - E[\mathbf{x}])(\mathbf{x} - E[\mathbf{x}])^T] \approx \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})(\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})^T = \mathbf{U}_{kl(d)} \mathbf{A}_{kl(d)} \mathbf{U}_{kl(d)}^T, \quad (5)$$

where $\mathbf{U}_{kl(d)} = [\mathbf{u}_{kl1} \ \mathbf{u}_{kl2} \ \cdots \ \mathbf{u}_{kld}]$ and $\mathbf{A}_{kl(d)} = \text{diag}([\lambda_{kl1} \ \lambda_{kl2} \ \cdots \ \lambda_{kld}]^T)$, respectively denote the corresponding eigenvector matrix and eigenvalue matrix of $\mathbf{C}_{\mathbf{x}|kl}$ with $\lambda_{kl1} \geq \lambda_{kl2} \geq \cdots \geq \lambda_{kld}$.

The covariance matrix's PCA-based subspace feature vector of $\mathbf{x}$ is

$$\mathbf{y} = \mathbf{U}_{kl(q)}^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl}) = [\mathbf{u}_{kl1} \ \mathbf{u}_{kl2} \ \cdots \ \mathbf{u}_{klq}]^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl}), \quad (6)$$

where $q < d$. Due to the Gaussian distribution assumption and the orthogonality between $\mathbf{u}_{klj}$ ($j = 1, 2, \dots, q$), y_j ($j = 1, 2, \dots, q$) should be independent and follow Gaussian distribution, namely $\mathbf{y} \sim \mathcal{N}(\mathbf{m}_{\mathbf{y}|kl}, \mathbf{C}_{\mathbf{y}|kl})$,

$$\mathbf{m}_{\mathbf{y}|kl} \approx \frac{1}{N} \sum_{i=1}^N \mathbf{U}_{kl(q)}^T (\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl}) = \mathbf{U}_{kl(q)}^T \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl} \right) \approx \mathbf{0}_{(q)}, \quad (7)$$

$$\mathbf{C}_{\mathbf{y}|kl} \approx \frac{1}{N} \sum_{i=1}^N \mathbf{U}_{kl(q)}^T (\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})(\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})^T \mathbf{U}_{kl(q)} \approx \mathbf{U}_{kl(q)}^T \mathbf{C}_{\mathbf{x}|kl} \mathbf{U}_{kl(q)} = \mathbf{U}_{kl(q)}^T (\mathbf{U}_{kl(d)} \mathbf{A}_{kl(d)} \mathbf{U}_{kl(d)}^T) \mathbf{U}_{kl(q)} = \mathbf{A}_{kl(q)}, \quad (8)$$

where $\mathbf{0}_{(q)}$ denoting a $q \times 1$ vector with zero elements. Then

$$\begin{aligned} p(\mathbf{y}|l; k) &= (2\pi)^{-(q/2)} |\mathbf{A}_{kl(q)}|^{-(1/2)} \exp[(\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl})^T \mathbf{U}_{kl(q)} \mathbf{A}_{kl(q)}^{-1} \mathbf{U}_{kl(q)}^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl})] \\ &= \prod_{j=1}^q \frac{1}{\sqrt{2\pi\lambda_{klj}}} \exp\left\{-\frac{[\mathbf{u}_{klj}^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl})]^2}{2\lambda_{klj}}\right\}. \end{aligned} \quad (9)$$

Thus, the statistical model for the covariance matrix's PCA-based subspace of each HRRP frame can be established. Obviously, this method utilizes the statistical characteristics of the PCA-based subspace in the covariance matrix, thus is referred to as the PCA covariance subspace method in this paper.

3. A novel statistical recognition method based on hypersphere model for HRRP-based RATR

3.1. Hyperspherical spread of HRRP samples

As discussed in Section 1, different from general Gaussian-distributed data, amplitude L_2 -normalized HRRP samples are applied to RATR, in other words, geometrically speaking, HRRP samples in RATR spread on a unit hypersphere. The correlation matrix of $\mathbf{X}_{kl} = \{\mathbf{x}_{kli} \in \mathbb{R}^d | i = 1, 2, \dots, N\}$ is

$$\mathbf{R}_{\mathbf{x}|kl} = E[\mathbf{x}\mathbf{x}^T] \approx \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} \mathbf{x}_{kli}^T = \mathbf{V}_{kl(d)} \mathbf{\Sigma}_{kl(d)} \mathbf{V}_{kl(d)}^T, \quad (10)$$

where $\mathbf{V}_{kl(d)} = [\mathbf{v}_{kl1} \ \mathbf{v}_{kl2} \ \cdots \ \mathbf{v}_{kld}]$ and $\mathbf{\Sigma}_{kl(d)} = \text{diag}([\varsigma_{kl1} \ \varsigma_{kl2} \ \cdots \ \varsigma_{kld}]^T)$, respectively, denote the corresponding eigenvector matrix and eigenvalue matrix of $\mathbf{R}_{\mathbf{x}|kl}$ with $\varsigma_{kl1} \geq \varsigma_{kl2} \geq \cdots \geq \varsigma_{kld}$. Mathematically

speaking, $\mathbf{v}_{kl1}$, the first eigenvector of $\mathbf{R}_{x|kl}$, satisfies

$$\mathbf{v}_{kl1} = \arg \min_{\mathbf{v}} \sum_{i=1}^N \left\| \mathbf{x}_{kli} - \frac{\langle \mathbf{x}_{kli}, \mathbf{v} \rangle}{\|\mathbf{x}_{kli}\|_2 \times \|\mathbf{v}\|_2} \times \mathbf{v} \right\|_2^2 = \arg \min_{\mathbf{v}} \sum_{i=1}^N \|\mathbf{x}_{kli} - \langle \mathbf{x}_{kli}, \mathbf{v} \rangle \times \mathbf{v}\|_2^2 \quad \text{such that } \|\mathbf{v}\|_2 = 1, \quad (11)$$

where $\|\cdot\|_2$ denotes the L_2 norm operation of a vector, $\langle \cdot, \cdot \rangle$ denotes the inner product operation between two vectors; while the average vector of this dataset satisfies

$$\mathbf{m}_{x|kl} = \arg \min_{\boldsymbol{\mu}} \sum_{i=1}^N \|\mathbf{x}_{kli} - \boldsymbol{\mu}\|_2^2. \quad (12)$$

Therefore, for a given HRRP frame $\mathbf{X}_{kl}$, its $\mathbf{m}_{x|kl}$ and $\mathbf{v}_{kl1}$ should be different, as shown in Fig. 2(a) and (b). Nevertheless, we can deduce that the average vector and the first eigenvector of the correlation matrix of a dataset will be similar, if and only if the dataset compactly assembles on the hypersphere. Under the limit condition that the dataset assembles so compactly that the hypersphere which the dataset assembles on turns into the tangent hyperplane, the average vector and the first eigenvector of the correlation matrix will be exactly same. Fig. 3(a) and (b) compares $\mathbf{m}_x$, the average vector, and $\mathbf{v}_1$, the first eigenvector of the correlation matrix, of a power transformed HRRP frame with $\alpha = 0.7$ in the measured data of “Yak-42”, which show the waveforms of the two vectors are similar. Therefore, an HRRP frame compactly assembles on the unit hypersphere. Moreover, Fig. 3(c) and (d) shows that the waveforms of $\mathbf{u}_1$, the first eigenvector of the covariance matrix, and $\mathbf{v}_2$, the second eigenvector of the correlation matrix, of this frame are also similar with $\lambda_1 \approx \zeta_2$. Similarly, experimental results show $\mathbf{u}_2 \approx \mathbf{v}_3$, $\lambda_2 \approx \zeta_3$, etc., and only the small eigenvalues and the corresponding eigenvectors have evident difference. The covariance matrix of an HRRP frame describes the spread situation of the difference vectors between the samples and the average vector, while the first eigenvector of the correlation matrix of the HRRP frame denotes the largest energy direction and the other eigenvectors describe the spread situation of the samples’ projection vectors into the subspace orthogonal to the first eigenvector, therefore, we can draw the conclusion that the average vector and the correlation matrix’s first eigenvector of an HRRP frame approximate to be in the same direction, and the PCA-based subspace in the covariance matrix approximates to be same as the subspace made up of other eigenvectors with larger eigenvalues in the correlation matrix from the experimental results.

Similar to Eq. (6), the correlation matrix’s PCA-based subspace feature vector of $\mathbf{x}$ is defined as

$$\mathbf{z} = \mathbf{V}_{kl(q)}^T \mathbf{x} = [\mathbf{v}_{kl1} \quad \mathbf{v}_{kl2} \quad \cdots \quad \mathbf{v}_{klq}]^T \mathbf{x}. \quad (13)$$

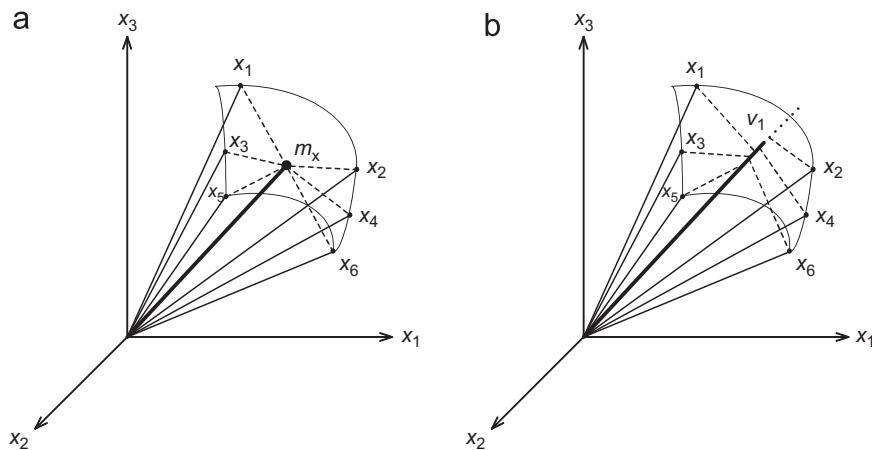


Fig. 2. Sketch maps: comparison between the average vector and the correlation matrix’s first eigenvector a dataset: (a) the average vector and (b) the first eigenvector of the correlation matrix.

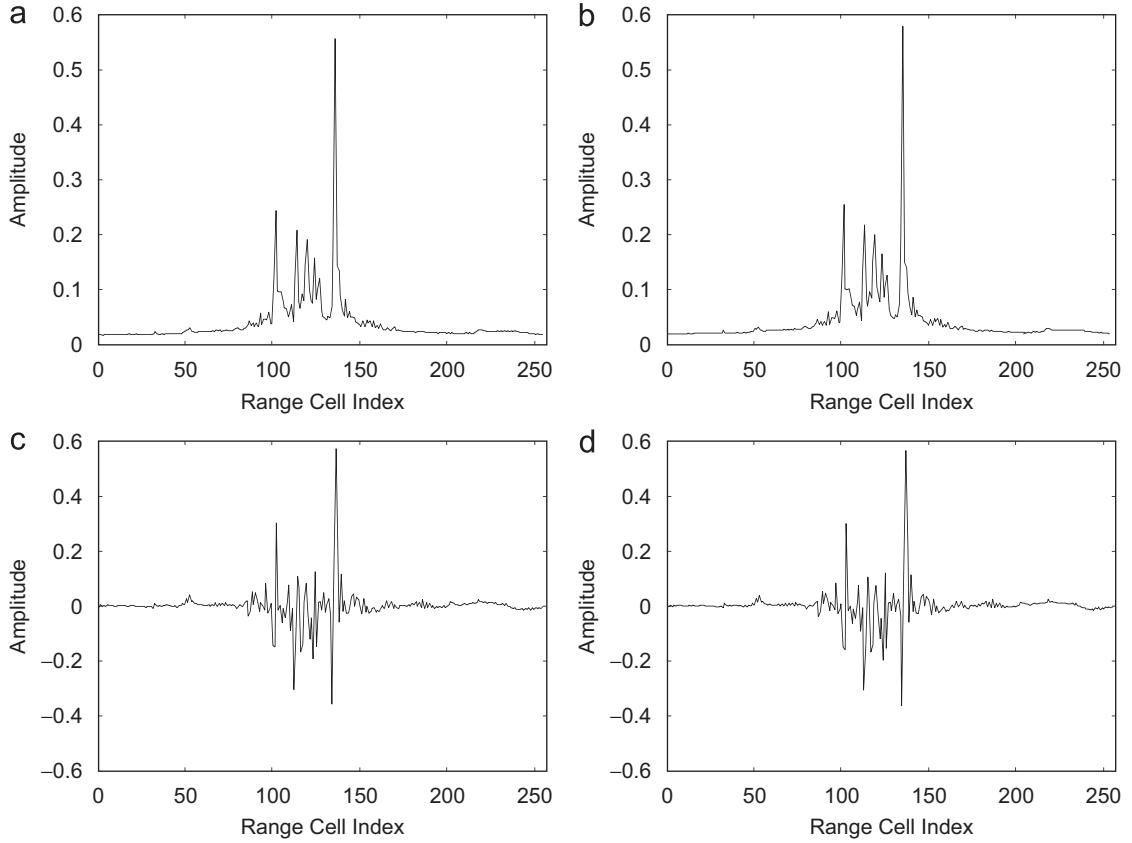


Fig. 3. Comparison between the covariance matrix and the correlation matrix of an HRRP frame with power parameter $\alpha = 0.7$ of “Yak-42”: (a) the average vector $\mathbf{m}_x$, (b) the first eigenvector of the correlation matrix $\mathbf{v}_1$, $\zeta_1 = 0.9219$, (c) the first eigenvector of the covariance matrix $\mathbf{u}_1$, $\lambda_1 = 0.0161$ and (d) the second eigenvector of the correlation matrix $\mathbf{v}_2$, $\zeta_2 = 0.0161$.

Due to the Gaussian distribution assumption and the orthogonality between $\mathbf{v}_{klj}$ ($j = 1, 2, \dots, q$), z_j ($j = 1, 2, \dots, q$) should be independent and follow Gaussian distribution, namely $\mathbf{z} \sim \mathcal{N}(\mathbf{m}_{\mathbf{z}|kl}, \mathbf{C}_{\mathbf{z}|kl})$, by formula derivation,

$$\mathbf{m}_{\mathbf{z}|kl} \approx \frac{1}{N} \sum_{i=1}^N \mathbf{V}_{kl(q)}^T \mathbf{x}_{kli} = \mathbf{V}_{kl(q)}^T \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} \right) \approx \mathbf{V}_{kl(q)}^T \mathbf{m}_{\mathbf{x}|kl}, \quad (14)$$

$$\mathbf{C}_{\mathbf{z}|kl} \approx \frac{1}{N} \sum_{i=1}^N \mathbf{V}_{kl(q)}^T (\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})(\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})^T \mathbf{V}_{kl(q)} \approx \mathbf{V}_{kl(q)}^T \mathbf{C}_{\mathbf{x}|kl} \mathbf{V}_{kl(q)}. \quad (15)$$

In the following, we will make a detailed analysis on the statistical characteristic of z_j ($j = 1, 2, \dots, q$), the element in the correlation matrix's PCA-based subspace feature vector, based on measured data. Fig. 4(a)–(f) shows the approximated probability density functions (pdfs) of some elements in the correlation matrix's PCA-based subspace feature vectors from the HRRP frame same as that used in Fig. 3, where the bar represents that approximated by histogram method, the dotted line represents that by Parzen windows using Gaussian kernels. According to the aforementioned analysis on hyperspherical spread of HRRP samples, the projection of each HRRP sample in a frame to the first eigenvector of the correlation matrix will be a little smaller than 1, and the mean of the projections of all HRRP samples in a frame to another eigenvector of the correlation matrix will approximate to 0, therefore, in theory, z_1 should follow a narrow distribution with nonzero mean, where $m_{z_1} \approx \mathbf{v}_1^T \mathbf{m}_x$ approximates to 1 and $C_{z_1} \approx \mathbf{v}_1^T \mathbf{C}_x \mathbf{v}_1 = \mathbf{v}_1^T (\mathbf{U}_d \Lambda_d \mathbf{U}_d^T) \mathbf{v}_1$ is very small with $m_{z_1}^2 + C_{z_1} \approx \zeta_1$, and z_j ($j = 2, 3, \dots, q$) should follow a distribution with zero mean, where $m_{z_j} \approx \mathbf{v}_j^T \mathbf{m}_x \approx 0$ and

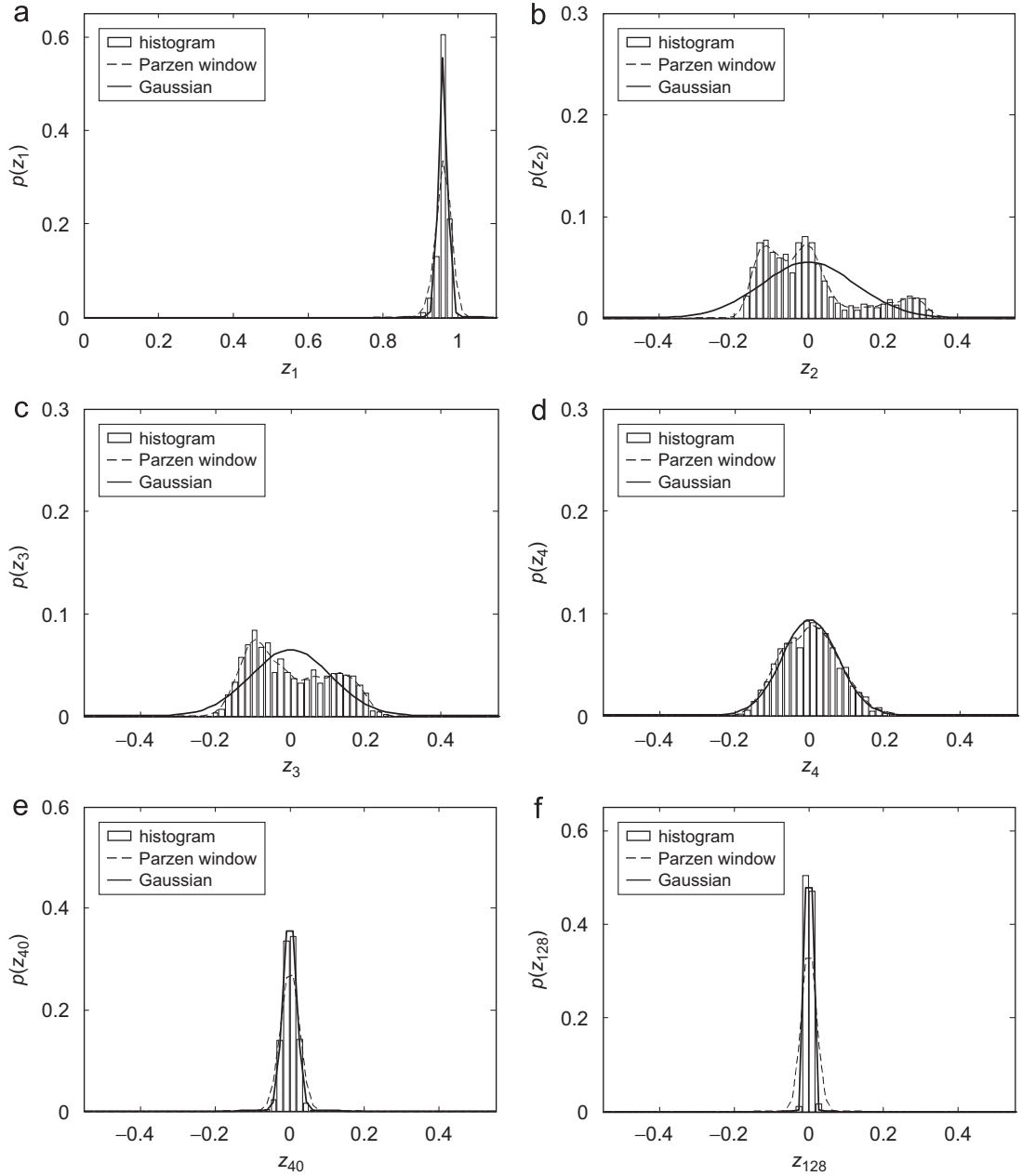


Fig. 4. The approximated pdfs of some elements in the correlation matrix's PCA-based subspace feature vector from the HRRP frame same as that used in Fig. 3: (a) z_1 $m_{z_1} = 0.9601$, $C_{z_1} = 1.57 \times 10^{-4}$, (b) z_2 , (c) z_3 , (d) z_4 , (e) z_{40} and (f) z_{128} .

$C_{z_j} \approx \mathbf{v}_j^T \mathbf{C}_x \mathbf{v}_j = \mathbf{v}_j^T (\mathbf{U}_d \mathbf{A}_d \mathbf{U}_d^T) \mathbf{v}_j \approx \lambda_{j-1} \approx \varsigma_j$. To simplify the statistical model, we assume that all of the elements in the feature vector are independent, and each element follows the corresponding Gaussian distribution. According to the experimental results shown in Fig. 4(a)–(f), this is true for z_j ($j = 1, 4, 5, \dots, q$), of which z_1 approximates to follow the Gaussian distribution $\mathcal{N}(m_{z_1}, C_{z_1})$ represented by solid line in Fig. 4(a), and z_j ($j = 4, 5, \dots, q$) approximates to follow the Gaussian distribution $\mathcal{N}(0, \varsigma_j)$ represented by solid line in Fig. 4(d)–(f). However, z_j ($j = 2, 3$) follows multimodal distribution, as shown in Fig. 4(b) and (c). This is caused by the fact that the power transformed HRRP samples do not follow jointly multivariate Gaussian distribution strictly. Nevertheless, our experimental results show that the feature elements evidently departing

from Gaussian distribution (usually 2–3) from power transformed HRRP samples are much less than those from original HRRP samples (usually 20–30). This further demonstrates that the power transformation with proper power parameters can correct for the departures from Gaussian distribution to some extent. Since after proper power transformation the feature elements not following Gaussian distribution are a few, we still assume that such a element is independent to other elements and follows Gaussian distribution with zero mean and variance equal to the corresponding eigenvalue of the correlation matrix, i.e. $z_j \sim \mathcal{N}(0, \varsigma_j)$, $j = 2, 3, \dots, q$. Therefore,

$$\mathbf{m}_{z_1|kl} \approx \begin{bmatrix} \mathbf{m}_{z_1|kl} \\ 0_{(q-1)} \end{bmatrix} \quad \text{and} \quad \mathbf{C}_{z_1|kl} \approx \begin{bmatrix} C_{z_1|kl} & \\ & \Sigma_{kl(2 \sim q)} \end{bmatrix},$$

then

$$\begin{aligned} p(z|l; k) &\approx (2\pi)^{-(q/2)} \left| \begin{bmatrix} C_{z_1|kl} & \\ & \Sigma_{kl(2 \sim q)} \end{bmatrix} \right|^{-(1/2)} \exp \left[\left(\mathbf{x} - \begin{bmatrix} \mathbf{m}_{z_1|kl} \\ \mathbf{0}_{(d-1)} \end{bmatrix} \right)^T \begin{bmatrix} C_{z_1|kl} & \\ & \Sigma_{kl(2 \sim q)} \end{bmatrix}^{-1} \left(\mathbf{x} - \begin{bmatrix} \mathbf{m}_{z_1|kl} \\ \mathbf{0}_{(d-1)} \end{bmatrix} \right) \right] \\ &= \frac{1}{\sqrt{2\pi C_{z_1|kl}}} \exp \left\{ -\frac{(\mathbf{v}_{kl1}^T \mathbf{x} - \mathbf{m}_{z_1|kl})^2}{2C_{z_1|kl}} \right\} \prod_{j=2}^q \frac{1}{\sqrt{2\pi \varsigma_{klj}}} \exp \left\{ -\frac{(\mathbf{v}_{klj}^T \mathbf{x})^2}{2\varsigma_{klj}} \right\}. \end{aligned} \quad (16)$$

Thus, the statistical model for the correlation matrix's PCA-based subspace of each HRRP frame can be established. Compared with the PCA covariance subspace method, this method utilizes the statistical characteristics of the PCA-based subspace in the correlation matrix, thus is referred to as the PCA correlation subspace method in this paper.

From the aforementioned analysis, we can see that the PCA correlation subspace is similar to the PCA covariance subspace except for the first eigenvector of the correlation matrix. In theory, an eigenvector or the combination of some eigenvectors of the covariance matrix can also denote the direction of the first eigenvector of the correlation matrix. Nevertheless, since $\mathbf{m}_x \approx \mathbf{v}_1$ and $\|\mathbf{m}_x\|_2 \approx 1$, these eigenvectors of the covariance matrix relate to small eigenvalues and are removed from the PCA covariance subspace. Actually, the distribution characteristics along this direction should be important in radar HRRP statistical recognition. Under the limit condition that $\mathbf{m}_{x|kl} = \mathbf{v}_{kl1}$, $\mathbf{v}_{klj} = \mathbf{u}_{klj-1}$, $\zeta_{klj} = \lambda_{klj-1}$ ($j = 2, 3, \dots, q$), by formula derivation we get that $z_1 \sim \delta(z_1 - 1)$ with

$$\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}, \quad z_j \sim \mathcal{N}(0, \varsigma_{klj}), \quad j = 2, 3, \dots, q,$$

in other words, one-dimensional feature vector z_1 is enough for statistical recognition. From this point of view, the PCA correlation subspace method should be more suitable for radar HRRP statistical recognition.

3.2. Signal subspace and noise subspace

In the aforementioned two recognition methods, the whole feature space is divided into the signal subspace and the noise subspace, and only the statistical characteristics in the signal subspace is utilized, which can not only denoise but also decrease computation complexity. However, is there any discriminating information in the noise subspace?

Actually, only a part of an HRRP sample includes target signal, referred to as the support region, and the other mainly includes noises. Although noises are uniformly distributed in a returned echo vector along the range cells, an HRRP sample is obtained after the demodulation, during which target signal can suppress the noises in the same range cells, therefore, the noises outside the support region are larger than those inside the support region for an HRRP sample. As shown in Figs. 3(b), (d) and 5(a)–(f), by comparing between the waveforms of eigenvectors of the correlation matrix of an HRRP frame, the eigenvectors with larger

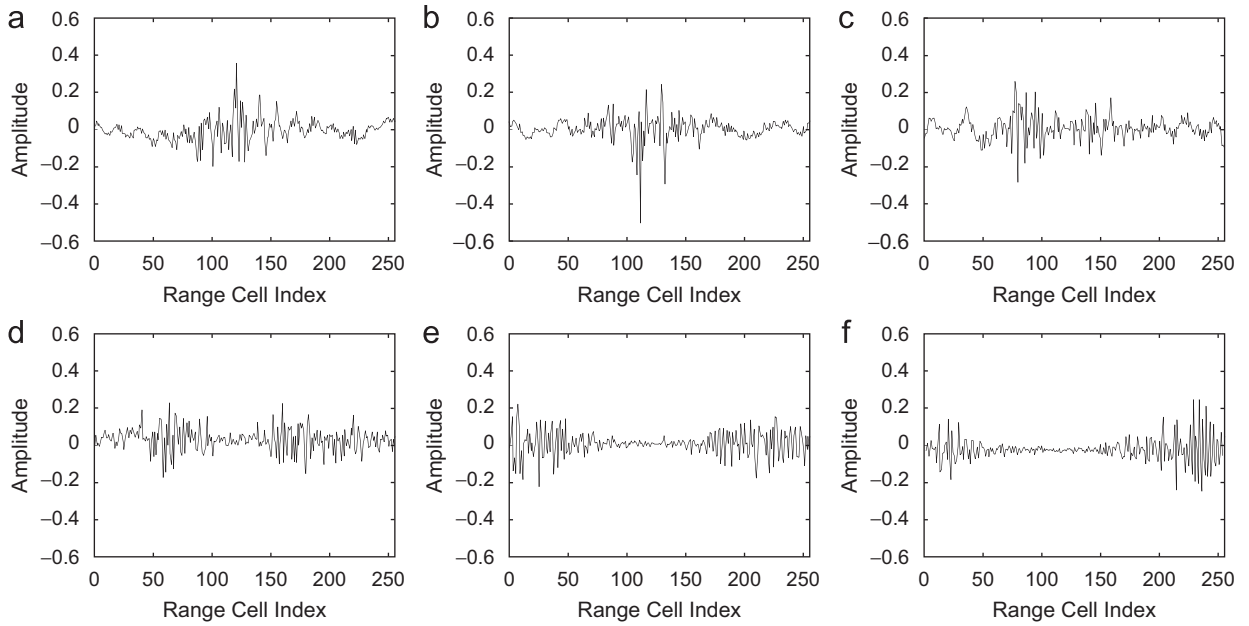


Fig. 5. Comparison between the eigenvectors of the correlation matrix of the HRRP frame same as that used in Figs. 3 and 4: (a) $\mathbf{v}_{10}$, $\zeta_{10} = 0.0014$, (b) $\mathbf{v}_{40}$, $\zeta_{40} = 2.97 \times 10^{-4}$, (c) $\mathbf{v}_{80}$, $\zeta_{80} = 1.30 \times 10^{-4}$, (d) $\mathbf{v}_{128}$, $\zeta_{128} = 6.16 \times 10^{-5}$, (e) $\mathbf{v}_{200}$, $\zeta_{200} = 2.36 \times 10^{-5}$ and (f) $\mathbf{v}_{256}$, $\zeta_{256} = 8.37 \times 10^{-6}$.

eigenvalues mainly describe the echoes inside the support region, while those with smaller eigenvalues mainly describe the echoes outside the support region. As shown in Fig. 5(d)–(f), the eigenvectors with very small eigenvalues completely describe the noises. Thus, the noise fluctuation information described by small eigenvalues and the corresponding eigenvectors is not useful for recognition, but the support region length information contained in the eigenvectors with small eigenvalues may be useful.

Similar to Eqs. (6) and (13), the correlation matrix's whole space feature vector of $\mathbf{x}$ is defined as

$$\mathbf{z} = \mathbf{V}_{(d)}^T \mathbf{x} = [\mathbf{v}_{kl1} \quad \mathbf{v}_{kl2} \quad \cdots \quad \mathbf{v}_{kld}]^T \mathbf{x}. \quad (17)$$

Due to the Gaussian distribution assumption and the orthogonality between $\mathbf{v}_{klj}$ ($j = 1, 2, \dots, d$), $\mathbf{z}_j$ ($j = 1, 2, \dots, d$) should be independent and follow Gaussian distribution. Obviously, $\mathbf{z}_j = z_j$ ($j = 1, 2, \dots, q$), $\mathbf{z}_{j'} \sim \mathcal{N}(m_{z_1}, C_{z_1})$, $\mathbf{z}_{j''} \sim \mathcal{N}(0, \zeta_{j''})$, $j' = 2, 3, \dots, q$. Let

$$\bar{\zeta}_{kl} = \frac{1}{(d-q)} \sum_{j=q+1}^d \zeta_{klj}. \quad (18)$$

Then we assume that the elements in the feature vector corresponding to the noise subspace of the correlation matrix are independent to other elements and each follows the Gaussian distribution with zero mean and the variance equal to the mean of the eigenvalues in the correlation matrix's noise subspace, i.e. $\mathbf{z}_{j''} \sim \mathcal{N}(0, \bar{\zeta}_{kl})$, $j'' = q+1, q+2, \dots, d$. Thus, the waveforms of the eigenvectors in the correlation matrix's noise subspace, which contain the support region length information, can be utilized; while the eigenvalues denoting the energies of the eigenvectors in the correlation matrix's noise subspace, which can describe the noise fluctuation information along with the eigenvectors, can be ignored. Therefore,

$$\mathbf{m}_{\mathbf{z}|kl} \approx \begin{bmatrix} m_{z_1|kl} \\ 0_{(d-1)} \end{bmatrix}, \quad \mathbf{C}_{\mathbf{z}|kl} \approx \begin{bmatrix} C_{z_1|kl} & & \\ & \Sigma_{kl(2 \sim q)} & \\ & & \bar{\zeta}_{kl} \mathbf{I}_{d-q} \end{bmatrix},$$

then

$$\begin{aligned}
 p(\mathbf{z}|l; k) &= (2\pi)^{-(d/2)} \left\| \begin{bmatrix} C_{z_1|kl} & & \\ & \Sigma_{kl(2\sim q)} & \\ & & \bar{\zeta}_{kl} \mathbf{I}_{d-q} \end{bmatrix} \right\|^{-(1/2)} \\
 &\quad \times \exp \left[\left(\mathbf{x} - \begin{bmatrix} \mathbf{m}_{z_1|kl} \\ \mathbf{0}_{(d-1)} \end{bmatrix} \right)^T \begin{bmatrix} C_{z_1|kl} & & \\ & \Sigma_{kl(2\sim q)} & \\ & & \bar{\zeta}_{kl} \mathbf{I}_{d-q} \end{bmatrix}^{-1} \left(\mathbf{x} - \begin{bmatrix} \mathbf{m}_{z_1|kl} \\ \mathbf{0}_{(d-1)} \end{bmatrix} \right) \right] \\
 &= \frac{1}{\sqrt{2\pi C_{z_1|kl}}} \exp \left\{ -\frac{(\mathbf{v}_{kl1}^T \mathbf{x} - m_{z_1|kl})^2}{2C_{z_1|kl}} \right\} \\
 &\quad \times \prod_{j=2}^q \frac{1}{\sqrt{2\pi \zeta_{klj}}} \exp \left\{ -\frac{(\mathbf{v}_{klj}^T \mathbf{x})^2}{2\zeta_{klj}} \right\} \prod_{j=q+1}^d \frac{1}{\sqrt{2\pi \bar{\zeta}_{kl}}} \exp \left\{ -\frac{(\mathbf{v}_{klj}^T \mathbf{x})^2}{2\bar{\zeta}_{kl}} \right\} \\
 &= p(\mathbf{z}|l; k) (2\pi \bar{\zeta}_{kl})^{-(d-q/2)} \exp \left\{ -\frac{1 - \sum_{j=1}^q (\mathbf{v}_{klj}^T \mathbf{x})^2}{2\bar{\zeta}_{kl}} \right\}. \tag{19}
 \end{aligned}$$

Although the aforementioned statistical model utilizes the statistical characteristics of the noise subspace, the computation complexity does not increase, due to the amplitude normalization, i.e. $\sum_{j=1}^d (\mathbf{v}_{klj}^T \mathbf{x})^2 = 1$. Compared with the PCA covariance subspace method and the PCA correlation subspace method, this method utilizes the statistical characteristics of the PCA-based whole space in the correlation matrix, thus is referred to as the PCA correlation space method in this paper. The experimental results will show this method can improve the recognition performance further.

4. Main procedure of the statistical recognition methods

4.1. Training phase

Step 1 : Frame division: Divide the training samples of target T_k ($k = 1, 2, \dots, K$) into HRRP frames $\{\mathbf{X}_{kl}\}_{l=1}^{L_k}$ with $\mathbf{X}_{kl} = \{\mathbf{x}_{kli} \in \mathbb{R}^d | i = 1, 2, \dots, N\}$, where $\mathbf{x}_{kli}$ denotes the i th HRRP sample in the l th frame of target T_k .

Step 2: Preprocessing: Firstly, apply power transformation or Box–Cox transformation with proper power parameter to each frame of training samples; secondly, compensate the time-shift sensitivity of each sample in each frame by the slide correlation processing or its modifications; thirdly, normalize the amplitude of each sample in each frame.

Step 3: Average profile: Estimate the average profile $\mathbf{m}_{\mathbf{x}|kl}$ by Eq. (4) for each frame.

Step 4: Parameter estimation: For the PCA covariance subspace method, parameters $\mathbf{U}_{kl(q)}$ and $\mathbf{A}_{kl(q)}$ can be estimated by Eq. (5); for the PCA correlation subspace method, parameters $m_{z_1|kl}$ can be estimated by Eq. (14), $C_{z_1|kl}$ by Eq. (15), $\mathbf{V}_{kl(q)}$ and $\Sigma_{kl(2\sim q)}$ by Eq. (10); for the PCA correlation space method, parameters $m_{z_1|kl}$, $C_{z_1|kl}$, $\mathbf{V}_{kl}$, $\Sigma_{kl(2\sim q)}$ can be estimated similar to in the PCA correlation subspace method, but $\bar{\zeta}_{kl}$ by Eq. (18).

Step 5: Frame template: Store both the average profile and the parameters for each frame of each target. For the PCA covariance subspace method, the frame template $\mathbb{M}_{\text{cov-sub}} = \{\mathbf{m}_{\mathbf{x}|kl}, \mathbf{U}_{kl(q)}, \mathbf{A}_{kl(q)}\}_{k=1, l=1}^{K, L_k}$; for the PCA correlation subspace method, the frame template $\mathbb{M}_{\text{cor-sub}} = \{\mathbf{m}_{\mathbf{x}|kl}, m_{z_1|kl}, C_{z_1|kl}, \mathbf{V}_{kl(q)}, \Sigma_{kl(2\sim q)}\}_{k=1, l=1}^{K, L_k}$; for the PCA correlation space method, the frame template $\mathbb{M}_{\text{cor}} = \{\mathbf{m}_{\mathbf{x}|kl}, m_{z_1|kl}, C_{z_1|kl}, \mathbf{V}_{kl}, \Sigma_{kl(2\sim q)}, \bar{\zeta}_{kl}\}_{k=1, l=1}^{K, L_k}$.

4.2. Test phase

Step 1: Preprocessing: Firstly, apply the power transformation or the Box–Cox transformation with same power parameter as that used for training samples to the test sample; secondly, normalize the amplitude of the test sample.

Step 2: Matching: Substitute the power transformed and amplitude normalized test sample $\mathbf{x}$ into the l th ($l = 1, 2, \dots, L_k$) frame model of target T_k ($k = 1, 2, \dots, K$). Firstly, match the test sample with the average profile by the slide correlation processing or its modifications to deal with the time-shift sensitivity problem; secondly, calculate the joint pdf of the feature vector of the time-shift compensated test sample by Eq. (9) for the PCA covariance subspace method, by Eq. (16) for the PCA correlation subspace method, or by Eq. (19) for the PCA correlation space method; thirdly, the class-conditional density of target T_k ($k = 1, 2, \dots, K$) is

$$p(\mathbf{x}|k) = \max_l p(\mathbf{x}|l; k) \quad (20)$$

fourthly, according to Bayes theorem, the posterior probability of target T_k ($k = 1, 2, \dots, K$) is

$$P(k|\mathbf{x}) = \frac{p(\mathbf{x}|k)P(k)}{\sum_{k=1}^K p(\mathbf{x}|k)P(k)} \quad (21)$$

where $P(k)$ is the prior class probability.

Step 3: Classification: If

$$J = \arg \max_k P(k|\mathbf{x}) \quad (22)$$

$\mathbf{x}$ belongs to target T_J .

From the algorithm procedure, we can know that the computation burden for the three statistical methods mainly depends on the eigenvalue decomposition of the autocovariance matrix or the autocorrelation matrix of each HRRP frame. Since the dimensionality of the autocorrelation matrix is same as that of the autocovariance matrix, our proposed methods do not increase the computational complexity.

5. Experimental result

5.1. Measured data

The results presented here are based on measured airplane data. The parameters of the targets and radar are shown in Table 1, and the projections of target trajectories onto ground plane are shown in Fig. 6, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig. 6, the measured data are segmented. Training data and test data are from different data segments, among which the second and the fifth segments of Yak-42, the sixth and the seventh segments of Cessna Citation S/II, the fifth and the sixth segments of An-26 are taken as the training data, other data segments are taken as the test data. As discussed in the Refs. [9–12,15–18,21], we need training data from different target-aspect angles to

Table 1
The parameters of planes and radar in the ISAR experiment

Radar parameters	Center frequency, 5520 MHz		
	Bandwidth, 400 MHz		
Planes	Length (m)	Width (m)	Height (m)
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

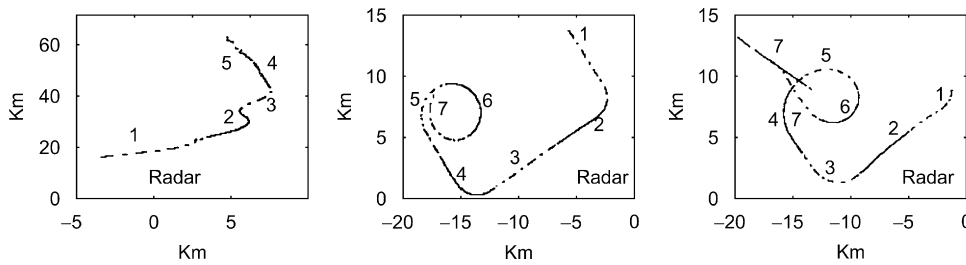


Fig. 6. The projections of target trajectories onto ground plane: (a) Yak-42, (b) Cessna Citation S/II and (c) An-26.

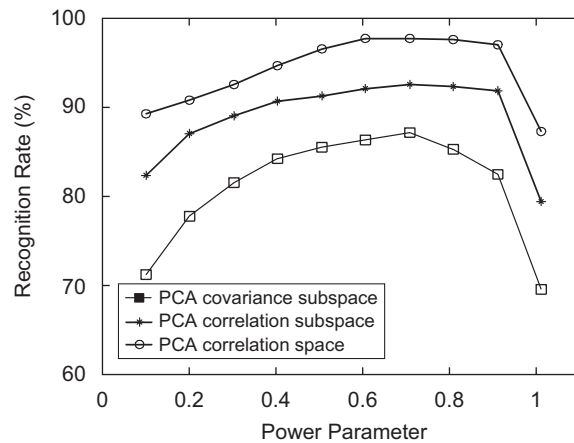


Fig. 7. The average recognition rates of the PCA covariance subspace method, the PCA correlation subspace method and the PCA correlation space method versus the power parameter ($q = 40$).

establish different frame models. This can be realized by using the simulated data generated by some electromagnetic simulation software, such as XPATCH, as the training data. However, for our measured data, we only require that the training data cover almost all of the target-aspect angles of the test data. According to Fig. 6, this can be satisfied. Moreover, the elevation angles of the test data are different from those of the training data, thus the generalization performance of the recognition methods can be tested.

5.2. Experimental results

The average recognition rates of the PCA covariance subspace method, the PCA correlation subspace method and the PCA correlation space method versus the power parameter α with $q = 40$ are shown in Fig. 7, of which the optimal power parameters for the three methods $\alpha^* = 0.7$. For a given power parameter and dimensionality parameter, the average recognition rate of the PCA correlation subspace method is approximately 5–11 percentage points larger than that of the PCA covariance subspace method, and that of the PCA correlation space method is further larger than that of the PCA correlation subspace method by approximately 4–8 percentage points without increasing computation complexity. Tables 2 and 3 show the confusion matrices and average recognition rates of the independent Gaussian model with the optimal power transformation ($\alpha^* = 0.5$) [6], the independent Gamma model [19,20], the independent two-distribution compounded statistical model [21], the PCA covariance subspace method with $q = 40$ and $\alpha^* = 0.7$, the PCA correlation subspace method with $q = 40$ and $\alpha^* = 0.7$, the PCA correlation space method with $q = 40$ and $\alpha^* = 0.7$ (The results in Table 2 are taken from Ref. [21]). In the six statistical methods, the PCA correlation space method is the best one, the independent two-distribution compounded statistical model is the next one, the average recognition rate of which is approximately 2 percentage points larger than that of the PCA

Table 2

The confusion matrices and average recognition rates of the independent Gaussian model with the optimal power transformation, the independent Gamma model and the independent two-distribution compounded statistical model

	The independent Gaussian model ($\alpha^* = 0.5$)			The independent Gamma model			The independent two-distribution compounded statistical model		
	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26
Yak-42	99.50	1.05	16.05	99.25	0.70	17.90	99.75	0	8.00
Cessna Citation S/II	0	96.00	9.95	0	89.50	1.40	0	93.60	1.60
An-26	0.50	2.95	74.00	0.75	9.80	80.70	0.25	6.40	90.40
Average recognition rates (%)	89.83			89.82			94.58		

Table 3

The confusion matrices and average recognition rates of the PCA covariance subspace method, the PCA correlation subspace method and the PCA correlation space method with $q = 40$ and optimal power parameters

	The PCA covariance subspace method ($\alpha^* = 0.7$)			The PCA correlation subspace method ($\alpha^* = 0.7$)			The PCA correlation space method ($\alpha^* = 0.7$)		
	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26
Yak-42	99.00	5.90	21.30	99.50	4.65	8.05	99.58	0.15	1.70
Cessna Citation S/II	0.58	89.70	6.20	0.17	90.50	4.40	0.25	99.00	3.90
An-26	0.42	4.40	72.50	0.33	4.85	87.55	0.17	0.85	94.40
Average recognition rates (%)	87.07			92.52			97.66		

correlation subspace method, and the PCA covariance subspace method is the last one, the average recognition rate of which is approximately 3 percentage points smaller than that of the independent Gaussian model. This shows that the statistical methods which consider the correlation between the elements in an HRRP sample but cannot describe the jointly statistical characteristics accurately also cannot achieve the good recognition performances.

6. Conclusion

There were some independent statistical models available for HRRP-based RATR. Nevertheless, theoretical analysis and our experimental results based on measured data show that the statistical independence assumption regarding the elements in an HRRP sample is not true. A novel statistical recognition method based on the PCA-based subspace in the correlation matrix of each HRRP frame is proposed for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis. Compared with the PCA covariance subspace statistical recognition method, the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace can be fully utilized in this method without increasing computation complexity. The experimental results based on measured data show that our proposed method can greatly improve the recognition performance.

It is worth pointing out that for radar HRRP statistical recognition, a large amount of training data are required to learn the statistical characteristics of target HRRP, therefore, the performance evaluation results will be sounder if more targets and more training data are concerned.

Acknowledgments

This work was partially supported by the National Science Foundation of China (No. 60772140) and the National Defense Advanced Research Foundation of China.

References

- [1] R.A. Mitchell, R. Dewall, Overview of high range resolution radar target identification, in: *Proceedings of Automatic Target Recognition Working Group*, Monterey, CA, 1994.
- [2] H.-J. Li, S.-H. Yang, Using range profiles as feature vectors to identify aerospace objects, *IEEE Trans. A.P.* 41 (3) (1993) 261–268.
- [3] S. Hudson, D. Psaltis, Correlation filters for aircraft identification from radar range profile, *IEEE Trans. A.E.S.* 29 (3) (1993) 1741–1748.
- [4] A. Zyweck, R.E. Bogner, Radar target classification of commercial aircraft, *IEEE Trans. A.E.S.* 32 (2) (1996) 598–606.
- [5] R.A. Mitchell, Robust high range resolution radar target identification using a statistic feature based classifier with feature level fusion, Ph.D. Dissertation, University of Dayton, 1997.
- [6] S.P. Jacobs, Automatic target recognition using high-resolution radar range profiles, Ph. D. Dissertation, Washington University, 1999.
- [7] R. Van der Heiden, F.C.A. Groen, The Box–Cox metric for nearest neighbour classification improvement, *Pattern Recog.* 30 (2) (1997) 273–279.
- [8] A.K. Shaw, R. Vagist, R. Williams, HRR-ATR using Eigen-templates with observation in unknown target scenario, *Proc. SPIE* 4053 (2000) 467–478.
- [9] R. Williams, J. Westerkamp, D. Gross, A. Palomino, Automatic target recognition of time critical moving targets 1D high range resolution (HRR) radar, *IEEE AES Syst. Mag.* 15 (4) (2000) 37–43.
- [10] X.-J. Liao, Z. Bao, M.-D. Xing, On the aspect sensitivity of high resolution range profiles and its reduction methods, *The Record of the IEEE 2000 International Radar Conference*, Alexandria, May 2000, pp. 310–315.
- [11] L. Du, H.-W. Liu, Z. Bao, Radar HRRP target recognition based on higher-order spectra, *IEEE Trans. S.P.* 53 (7) (2005) 2359–2368.
- [12] K.T. Kim, D.K. Seo, H.T. Kim, Efficient radar target recognition using the MUSIC algorithm and invariant feature, *IEEE Trans. A.P.* 50 (3) (2002) 325–337.
- [13] K. Rosenbach, J. Schiller, Non-co-operative air target identification using radar imagery: identification rate as a function of signal bandwidth, *The Record of the IEEE 2000 International Radar Conference*, Alexandria, May 2000, pp. 305–309.
- [14] D.E. Nelson, J.A. Starzyk, D.D. Ensley, Iterated wavelet transformation and signal discrimination for HRR radar target recognition, *IEEE Trans. SYST. MAN. CY. A* 33 (1) (2002) 52–57.
- [15] X.-J. Liao, P. Runkle, L. Carin, Identification of ground targets from sequential high-range-resolution radar signatures, *IEEE Trans. A.E.S.* 38 (4) (2002) 1230–1242.
- [16] H.-W. Liu, Z. Bao, Radar HRR Profiles recognition based on SVM with power-transformed-correlation kernel, *Springer Lecture Notes in Computer Science*, Springer, Berlin, August 2004, pp. 531–536.
- [17] Y. Shi, X.-D. Zhang, A gabor atom network for signal classification with application in radar target recognition, *IEEE Trans. S.P.* 49 (12) (2001) 2994–3004.
- [18] J. Zwart, R. Heiden, S. Gelsema, F. Groen, Fast translation invariant classification of HRR range profiles in A zero phase representation, *IEE Proc.-Radar Sonar Navig.* 150 (6) (2003) 411–418.
- [19] A.R. Webb, Gamma mixture models for target recognition, *Pattern Recog.* 33 (2000) 2045–2054.
- [20] K. Copesey, A.R. Webb, Bayesian gamma mixture model approach to radar target recognition, *IEEE Trans. A.E.S.* 39 (4) (2003) 1201–1217.
- [21] L. Du, H.-W. Liu, Z. Bao, J.-Y. Zhang, A two-distribution compounded statistical model for radar HRRP target recognition, *IEEE Trans. S.P.* 54 (6) (2006) 2226–2238.
- [22] W.G. Carrara, R.S. Goodman, R.M. Majewski, *Spotlight Synthetic Aperture Radar—Signal Processing Algorithms*, Artech House, Norwood, MA, 1995.
- [23] W. Ye, Study of the inverse synthetic aperture radar imaging and motion compensation, Ph.D. Dissertation, Xidian University, 1996.
- [24] K. Fukunaga, *Introduction to Statistical Pattern Recognition*, second ed., Academic, Boston, 1990.
- [25] R.M. Sakia, The Box–Cox Transformation Technique: A Review, *Statiscian* 41 (1992) 169–178.